

Student's 't' Statistics:-

and it follows a distribution
Known Student t-distribution
with pdf

$$f(t) = \frac{1}{\sqrt{\theta} B(\frac{1}{2}, \theta/2)} \frac{1}{(1 + \frac{t^2}{\theta})^{(\theta+1)/2}}$$

$$-\infty < t < \infty.$$

where $\theta = (n-1)$ ~~df~~ is df.

If $\theta = 1$ then

$$f(t) = \frac{1}{B(\frac{1}{2}, \frac{1}{2})} \frac{1}{(1+t^2)}$$

$$= \frac{1}{\pi} \frac{1}{(1+t^2)}$$

which is standard Cauchy's distribution.

$f(t)$ is symmetrical about the line $t=0$.

$\therefore$ odd order ~~moments~~ raw moment

T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31

is zero $\therefore$ mean $\mu_1 = 0$.

$$\mu_2' = \int_{-\infty}^{\infty} t^2 f(t) dt$$

$$= \frac{n^2 (n-1)(n-3) \dots 3 \cdot 1}{(n-2)(n-4) \dots (n-2r)}$$

$$\therefore \sigma^2 \equiv \text{Variance} = \mu_2 = \frac{n}{n-2}$$

mgf of t-distribution does not exist.

